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AIM:Decision Tree Learning

- Implement the Decision Tree Learning algorithm to build a decision tree for
- a given dataset.
- Evaluate the accuracy and effectiveness of the decision tree on test data.
- Visualize and interpret the generated decision tree.

CODE:

```
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree


df = pd.read_csv("C:/Users/hrane/Downloads/online food delivery_107.csv")

print("First 5 rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)


df.columns = df.columns.str.strip()
df = df.apply(
    lambda x: x.str.strip() if x.dtype == "object" else x
)


df["Output"] = df["Output"].map({"Yes": 1, "No": 0})


X = df.drop(["Output", "Feedback"], axis=1, errors="ignore")
y = df["Output"]


X = pd.get_dummies(X, drop_first=True)

print("\nNumber of Features after One-Hot Encoding:", X.shape[1])
print("Number of Samples:", X.shape[0])


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print("\nTraining Samples:", X_train.shape[0])
```

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```
print("Testing Samples:", X_test.shape[0])


model = DecisionTreeClassifier(
    criterion="entropy", max_depth=4, random_state=42
)

model.fit(X_train, y_train)
print("\nDecision Tree trained successfully!")


y_pred = model.predict(X_test)

print("\nPredicted Values:")
print(y_pred)

accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")

print("\nClassification Report:")
print(
    classification_report(
        y_test, y_pred, target_names=["No (0)", "Yes (1)"]
    )
)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

plt.figure(figsize=(25, 12))

plot_tree(
    model,
    feature_names=list(X.columns),
    class_names=["No", "Yes"],
    filled=True,
    rounded=True,
    fontsize=10,
)

plt.title("Decision Tree - Customer Output Classification")
plt.tight_layout()
plt.show()
```

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OUTPUT:

```
========== RESTART: C:\Users\hrane\Documents\AI_R.py ==========
First 5 rows:
   Age  Gender Marital Status  ... Output   Feedback Unnamed: 13
0   20  Female         Single  ...    Yes   Positive          Yes
1   24  Female         Single  ...    Yes   Positive          Yes
2   22    Male         Single  ...    Yes  Negative          Yes
3   22  Female         Single  ...    Yes   Positive          Yes
4   22    Male         Single  ...    Yes   Positive          Yes

[5 rows x 14 columns]

Dataset Shape:
(388, 14)

Number of Features after One-Hot Encoding: 22
Number of Samples: 388

Training Samples: 310
Testing Samples: 78

Decision Tree trained successfully!

Predicted Values:
[1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 0 1 1 1 1 1 1 1 0 1 1 1 1 0 1 1
 1 1 1 0 0 1 0 0 1 1 1 1 1 0 1 1 1 1 0 1 1 1 1 0 1 1 1 0 1 0 1 1 1 0 1 1 1 1 1 1 1 0 1
 1 1 1 0]

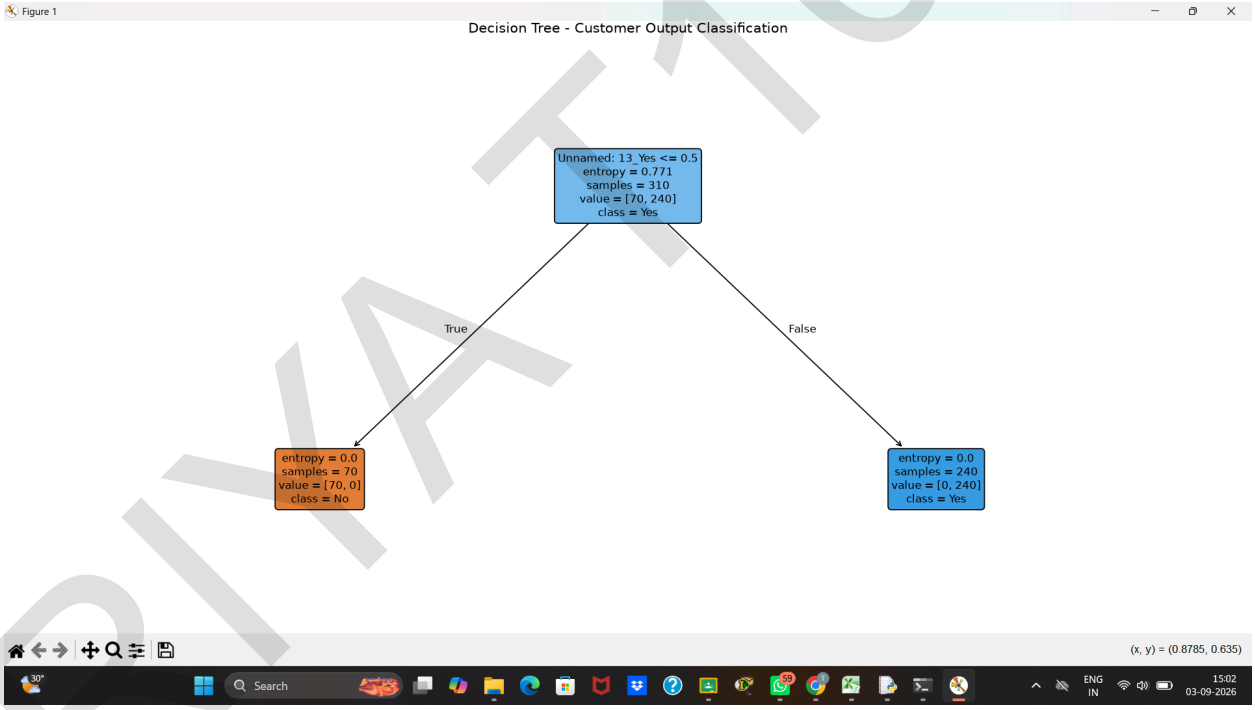
Accuracy: 1.0
Accuracy Percentage: 100.0 %

Classification Report:
              precision    recall  f1-score   support

      No (0)       1.00      1.00      1.00        17
     Yes (1)       1.00      1.00      1.00        61

    accuracy                           1.00        78
   macro avg       1.00      1.00      1.00        78
weighted avg       1.00      1.00      1.00        78


Confusion Matrix:
[[17  0]
 [ 0 61]]
```



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